1. Award: **0.83 points**

How much will $27,000 grow to in five years, assuming an interest rate of 9% compounded annually?
Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), and PVA of $1 (opens in a new tab)).

- $39,150
- $44,548
- → $41,543
- $68,543

FV = $27,000 × 1.53862 (Table FV of $1; n = 5; i = 9%) = $41,543

References

Multiple Choice Difficulty: 2 Medium Learning Objective: 05-02 Compute the future value of a single amount.

2. Award: 0.83 points

Today, Thomas deposited $200,000 in a three-year, 4% investment account that compounds quarterly. What is the maturity value of the investment account?

Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), and PVA of $1 (opens in a new tab)).

- $223,966
- $224,973
- $224,000
- → $225,366

FV = $200,000 × 1.12683 (Table FV of $1; n = 12; i = 1%) = $225,366

References

Multiple Choice	Difficulty: 2 Medium	Learning Objective: 05-02 Compute the future value of a single amount.

3. Award: **0.83 points**

An investment product promises to pay $56,000 at the end of 10 years. If an investor feels this investment should produce a rate of return of 11%, compounded annually, what's the most the investor should be willing to pay for the investment?

Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), PVA of $1 (opens in a new tab), FVAD of $1 (opens in a new tab) and PVAD of $1 (opens in a new tab))

- → $19,722
- $56,000
- $159,008
- $21,892

$56,000 × 0.35218* = $19,722 (rounded)
*PV of $1: $n = 10$; $i = 11\%$

References

Multiple Choice Difficulty: 2 Medium Learning Objective: 05-03 Compute the present value of a single amount.

4. Award: 0.83 points

What is the value today of receiving $3,700 at the end of four years, assuming an interest rate of 7% compounded annually?

Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), and PVA of $1 (opens in a new tab)).

- $2,664
- $3,223
- → $2,823
- $2,923

PV = $3,700 × 0.76290 (PV of $1; n = 4; i = 7%) = $2,823

References

Multiple Choice	Difficulty: 2 Medium	Learning Objective: 05-03 Compute the present value of a single amount.

5. Award: **0.83 points**

Davenport Incorporated offers a new employee two options. First, the employee can receive a one-time signing bonus at the date of employment. Second, the employee can take $31,000 at the date of employment and another $50,000 three years later. Assuming the employee's time value of money is 9% annually, what single payment in the first option would be equal to the total of the payments in the second option?

Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), and PVA of $1 (opens in a new tab)).

- $73,938
- $81,000
- $30,500
- → $69,609

The one-time equivalent would be $31,000 + the present value of $50,000; where n = 3 and i = 9%. That is, $31,000 + ($50,000 × 0.77218 from Table PV of $1) = $69,609

References

Multiple Choice	Difficulty: 3 Hard	Learning Objective: 05-03 Compute the present value of a single amount.

6. Award: **0.83 points**

At the end of each of the next four years, a new machine is expected to generate net cash flows of $11,500, $13,500, $12,500, and $18,000, respectively. What are the cash flows worth today if a 8% interest rate properly reflects the time value of money in this situation?

Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), and PVA of $1 (opens in a new tab)).

- → $45,376
- $32,145
- $46,576
- $55,500

PV = ($11,500 × 0.92593) + ($13,500 × 0.85734) + ($12,500 × 0.79383) + ($18,000 × 0.73503) = $45,376. Factors are from PV of $1.

References

Multiple Choice	Difficulty: 3 Hard	Learning Objective: 05-03 Compute the present value of a single amount.

7. Award: 0.83 points

How much will $5,000 invested at the end of each year grow to in seven years, assuming an interest rate of 11% compounded annually?

Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), and PVA of $1 (opens in a new tab)).

- $38,500
- → $48,917
- $31,142
- $78,918

FVA = $5,000 × 9.7833 (FVA of $1; n = 7; i = 11%) = $48,917

References

Multiple Choice	Difficulty: 2 Medium	Learning Objective: 05-07 Compute the future value of both an ordinary annuity and an annuity due.

8. Award: 0.83 points

How much will $3,500 invested at the end of each year grow to in 3 years, assuming an interest rate of 8% compounded annually?

Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), and PVA of $1 (opens in a new tab)).

- $9,962
- $8,400
- → $11,362
- $13,227

FVA = $3,500 × 3.2464 (FVA of $1; n = 3; i = 8%) = $11,362

References

Multiple Choice	Difficulty: 2 Medium	Learning Objective: 05-07 Compute the future value of both an ordinary annuity and an annuity due.

9. Award: 0.83 points

Claudine Corporation will deposit $5,000 into a money market account at the end of each year for the next four years. How much will accumulate by the end of the fourth and final payment if the account earns 8% interest? **Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar.** (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), and PVA of $1 (opens in a new tab)).

- $21,781
- → $22,531
- $20,000
- $20,300

FVA = $5,000 × 4.5061 (FVA of $1; n = 4; i = 8%) = $22,531

References

Multiple Choice	Difficulty: 2 Medium	Learning Objective: 05-07 Compute the future value of both an ordinary annuity and an annuity due.

10. Award: **0.83 points**

Quaker State Incorporated offers a new employee two options. First, the employee can receive a one-time signing bonus at the date of employment. Second, the employee can take $7,500 at the date of employment plus $22,000 at the end of each of his first three years of service. Assuming the employee's time value of money is 11% annually, what single payment in the first option would be equal to the total of the payments in the second option?
Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), and PVA of $1 (opens in a new tab)).

- $65,262
- → $61,262
- $72,089
- None of these answer choices are correct.

The one-time equivalent would be $7,500 + The present value of a $22,000 ordinary annuity; where n = 3 and i = 11%.

That is, $7,500 + ($22,000 × 2.44371 from Table PVA of $1) = $61,262.

References

Multiple Choice	Difficulty: 3 Hard	Learning Objective: 05-08 Compute the present value of an ordinary annuity, an annuity due, and a deferred annuity.

11. Award: 0.83 points

What is the value today of receiving $5,000 at the end of each year for the next four years, assuming an interest rate of 10% compounded annually?

Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), and PVA of $1 (opens in a new tab)).

- $15,649
- $23,205
- → $15,849
- $20,000

PVA = $5,000 × 3.16987 (PVA of $1; n = 4; i = 10%) = $15,849

References

Multiple Choice	Difficulty: 2 Medium	Learning Objective: 05-08 Compute the present value of an ordinary annuity, an annuity due, and a deferred annuity.

12. Award: **0.83 points**

Mary Alice just won the lottery and is trying to decide between the options of receiving the annual cash flow payment option of $260,000 per year for 30 years beginning today, or receiving one lump-sum amount today. Mary Alice can earn 5% investing this money. At what lump-sum payment amount would she be indifferent between the two alternatives?

Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), PVA of $1 (opens in a new tab), FVAD of $1 (opens in a new tab) and PVAD of $1 (opens in a new tab))

- $7,800,000
- $3,996,837
- $4,456,678
- → $4,196,678

$260,000 × 16.14107* = $4,196,678
*PVAD of $1: n = 30; i = 5%

References

Multiple Choice	Difficulty: 3 Hard	Learning Objective: 05-08 Compute the present value of an ordinary annuity, an annuity due, and a deferred annuity.

13. Award: **0.83 points**

Miller borrows $360,000 to be paid off in two years. The loan payments are semiannual with the first payment due in six months, and interest is at 10%. What is the amount of each payment?

Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), and PVA of $1 (opens in a new tab)).

- $72,000
- $62,479
- $117,521
- → $101,524

$360,000 ÷ 3.54595 (Table PVA of $1; n = 4; i = 5%) = $101,524

References

Multiple Choice	Difficulty: 3 Hard	Learning Objective: 05-09 Solve for unknown values in annuity situations involving present value.

14. Award: 0.83 points

You borrow $37,000 to buy a boat. The loan is to be paid off in monthly installments over one year at 12% interest annually. The first payment is due one month from today. What is the amount of each monthly payment?
Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), PVA of $1 (opens in a new tab), FVAD of $1 (opens in a new tab) and PVAD of $1 (opens in a new tab))

- $3,083
- $3,039
- → $3,287
- None of the other answer choices are correct.

$37,000 ÷ 11.25508* = $3,287 (rounded)
*PVA of $1: $n = 12$; $i = 1.00\%$

References

Multiple Choice | Difficulty: 3 Hard | Learning Objective: 05-09 Solve for unknown values in annuity situations involving present value.

15. Award: 0.83 points

An investor purchases a 7-year, $1,000 par value bond that pays semiannual interest of $14. If the semiannual market rate of interest is 8%, what is the current market value of the bond?
Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), PVA of $1 (opens in a new tab), FVAD of $1 (opens in a new tab) and PVAD of $1 (opens in a new tab))

- → $456
- $485
- $1,000
- $1,115

$14 × 8.2442*	=	$ 115
$1,000 × 0.3405**	=	340
		$ 456

*PVA of $1: n = 14; i = 8%
**PV of $1: n = 14; i = 8%

References

Multiple Choice | Difficulty: 3 Hard | Learning Objective: 05-10 Briefly describe how the concept of the time value of money is incorporated into the valuation of bonds, long-term leases, installment notes, and pension obligations.

16. Award: **0.83 points**

On January 1, 2027, Glanville Company sold goods to Otter Corporation. Otter signed an installment note requiring payment of $18,000 annually for six years. The first payment was made on January 1, 2027. The prevailing rate of interest for this type of note at date of issuance was 10%. What is the amount of sales revenue that should be recorded by Glanville in January 2027?

Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), PVA of $1 (opens in a new tab), FVAD of $1 (opens in a new tab) and PVAD of $1 (opens in a new tab))

- $108,000
- $78,395
- → $86,234
- None of the other answer choices are correct.

$18,000 × 4.79079* = $86,234 (rounded)
*PVAD of $1: n = 6; i = 10%

References

Multiple Choice	Difficulty: 2 Medium	Learning Objective: 05-10 Briefly describe how the concept of the time value of money is incorporated into the valuation of bonds, long-term leases, installment notes, and pension obligations.

17. Award: **0.83 points**

On January 1, 2027, Glanville Company sold goods to Otter Corporation. Otter signed an installment note requiring payment of $16,650 annually for six years. The first payment was made on January 1, 2027. The prevailing rate of interest for this type of note at date of issuance was 7%. What is the amount of the note receivable that should be recorded by Glanville in January 2027?

Note: Use tables, Excel, or a financial calculator. Round your final answer to the nearest whole dollar. (FV of $1 (opens in a new tab), PV of $1 (opens in a new tab), FVA of $1 (opens in a new tab), PVA of $1 (opens in a new tab), FVAD of $1 (opens in a new tab) and PVAD of $1 (opens in a new tab))

- $83,250
- $84,918
- → $68,268
- $73,610

$16,650 × 4.10020* = $68,268 (rounded)
*PVA of $1: *n* = 5; *i* = 7%

References

Multiple Choice	Difficulty: 2 Medium	Learning Objective: 05-10 Briefly describe how the concept of the time value of money is incorporated into the valuation of bonds, long-term leases, installment notes, and pension obligations.